# Demo: Loading Macros via `include-in-header`

This page demonstrates the `include-in-header` (and `include-before-body`) strategy for loading macros. The YAML front matter of this file includes:

```yaml
format:
  html:
    include-in-header:
      - macros-header.html   # HTML file wrapping macros in $$...$$ for MathJax
  pdf:
    include-in-header:
      - macros.qmd           # raw LaTeX goes directly into preamble
  revealjs:
    output-file: demo-include-in-header-slides.html
    include-before-body:
      - macros-header.html   # placed in <body> so MathJax processes it
  docx: default
```

> **Note (HTML):** For HTML output, `include-in-header` inserts content into the page `<head>`. The file `macros-header.html` wraps the macro definitions in `<div style="display:none">$$...$$</div>`, which the browser moves into the `<body>` and MathJax then processes.
>
> **Note (RevealJS):** For RevealJS, `include-before-body` is used instead of `include-in-header`. RevealJS processes MathJax per-slide, and macro definitions must be in the `<body>` — before the RevealJS presentation container — to be registered globally before any slide math is rendered. `include-before-body` places the hidden div at the very start of `<body>`. View the RevealJS slides output.
>
> **Note (PDF):** For PDF, `macros.qmd` can be included directly since its raw LaTeX goes into the document preamble.
>
> **Note:** This demo uses `macros-header.html` / `macros.qmd` because both files are at the root of this repository. When using this repository as a Git submodule (cloned into a `macros/` subdirectory), replace with `macros/macros-header.html` and `macros/macros.qmd`.

The same configuration can be placed in a shared `_quarto.yml` to apply project-wide:

```yaml
format:
  html:
    include-in-header:
      - macros/macros-header.html
  pdf:
    include-in-header:
      - macros/macros.qmd
  revealjs:
    include-before-body:
      - macros/macros-header.html
```

This inserts the macro definitions for each output format:

- **PDF**: `macros.qmd` goes into the LaTeX preamble via `include-in-header` — all macros work, including `\providecommand`.
- **HTML**: `macros-header.html` is inserted via `include-in-header` — the browser moves the `<div>` to the body and MathJax processes the `$$...$$` block. MathJax-supported commands (`\def`, `\newcommand`, `\renewcommand`, `\let`, environments) are recognized; `\providecommand` is silently ignored.
- **RevealJS**: `macros-header.html` is inserted via `include-before-body` (not `include-in-header`) — this places the hidden div at the start of `<body>` where RevealJS's MathJax will pick it up before rendering slides. The same MathJax limitations apply.
- **docx**: Word documents do not use a LaTeX preamble or MathJax header, so `include-in-header` has no effect on math rendering for docx. The `docx: default` entry above produces docx output without macro support.

### Example math

The table below uses macros defined with `\def` in `macros.qmd`, which work in all output formats that support MathJax or LaTeX:

| Description | Code | Output |
|---|---|---|
| Greek: alpha | `$\a$` | $\alpha$ |
| Greek: beta | `$\b$` | _ |
| Greek: gamma | `$\g$` | $\gamma$ |
| Greek: lambda | `$\l$` | |
| Greek: sigma | `$\s$` | $\sigma$ |
| Log-likelihood | `$\llik$` | $\ell$ |
| Independent | `$X \ind Y$` | $X \perp\!\!\!\perp Y$ |

| Description | Code | Output |
| --- | --- | --- |
| IID distributed | `$X_i \siid \Normal(\m, \ss)$` | $X_i \ \sim_{\text{iid}}\ \mathrm{N}(\mu, )$ |
| Vector: mu | `$\vm$` | $\vec{\mu}$ |
| Hat beta | `$\hb$` | $\hat{\beta}$ |